

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

Announcements

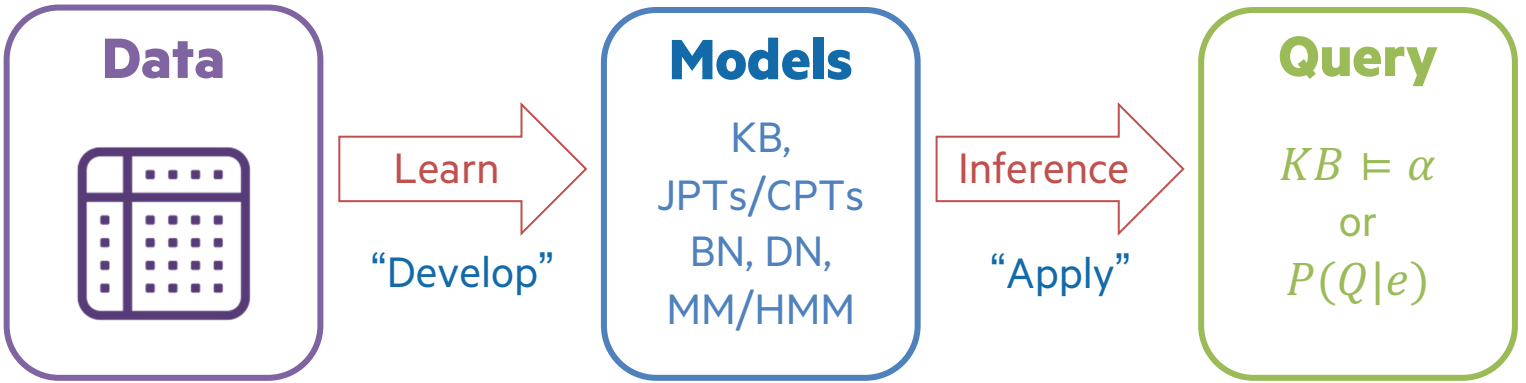
- Classwork 5
  - Due on Friday, 11/8 @5pm
- Homework 3
  - Due on Friday, 11/15 @5pm
  - Configure Python 2.7.x environment

This Lecture

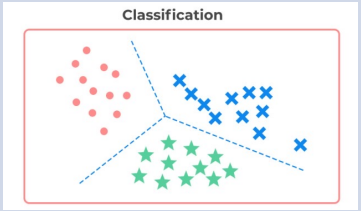
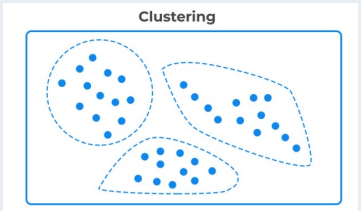
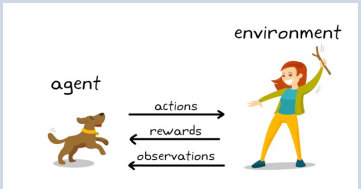
- Naïve Bayes Classification
  - Structure and Semantics
  - Decision Rule
  - Example and Practice

Last Lecture

- Machine Learning
  - Computationally discovering/constructing a model from data/examples/experience.
- Learning vs. Inference

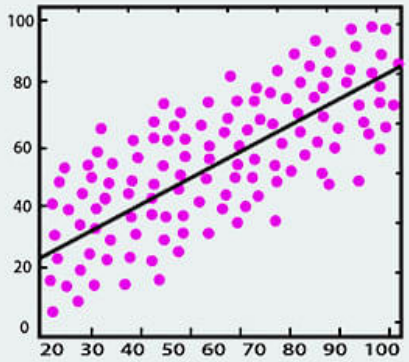


Types of Learning

Type	Feedback	Data	Goal	Example
Supervised Learning	Immediate & explicit	Labeled (input-output)	Find mapping between inputs and outputs	
Unsupervised Learning	No feedback	Unlabeled	Find structure / patterns in data	
Reinforcement Learning	Delayed & reward-based	No predefined data, but need rewards	Maximize rewards	

(Supervised) Regression vs. Classification

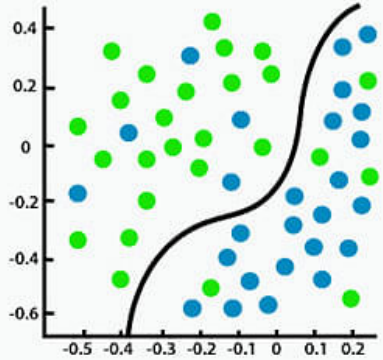
Output:
continuous numerical values



Regression

versus

Output:
discrete categories/labels



Classification

Example – Email Classification

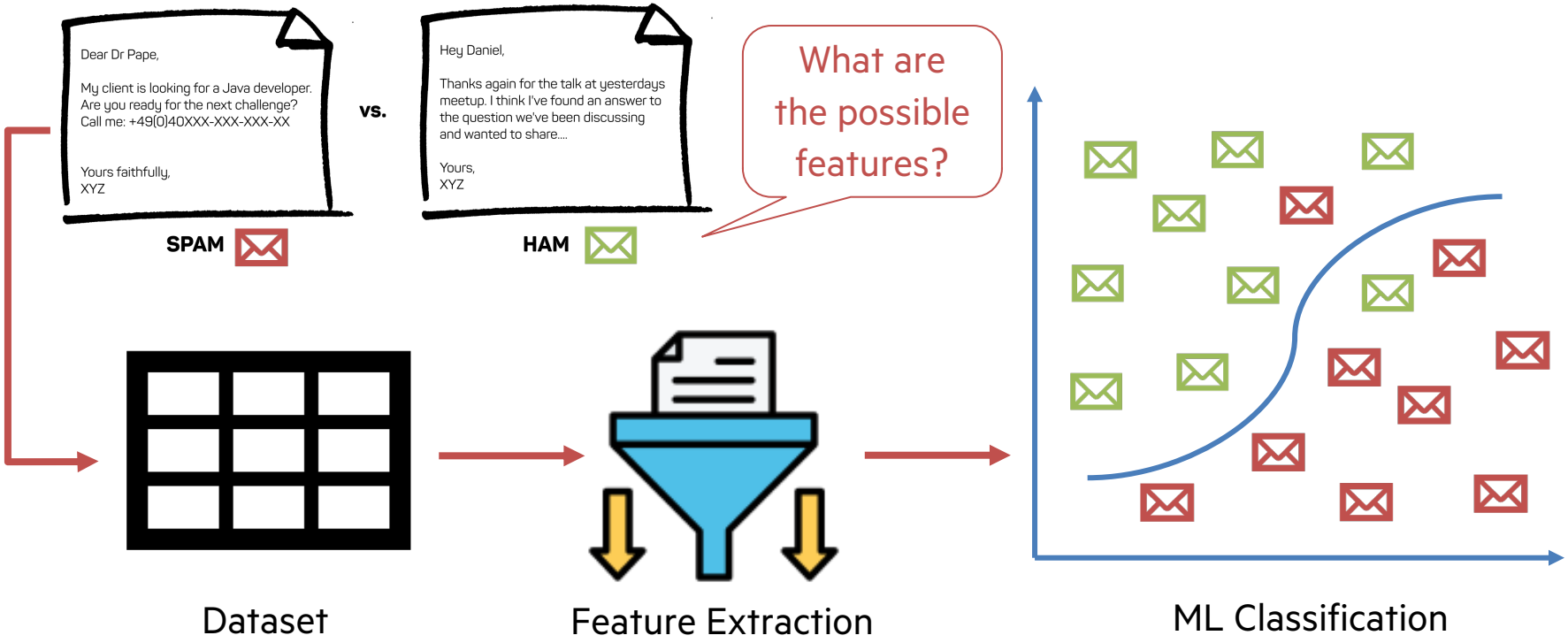


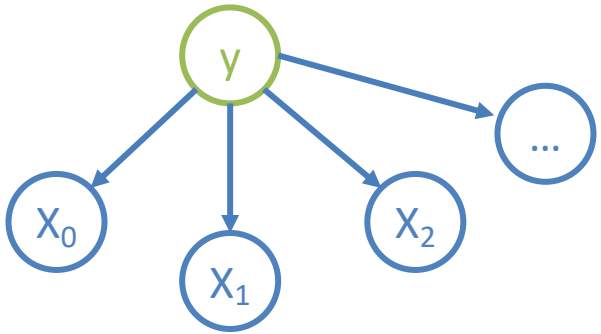
Image: https://www.codecentric.de/wissens-hub/blog/spam-classification-using-sparks-dataframes-ml-zeppelin-part-1

Naïve Bayes Classification

- Models the association between **features** X and **class** Y .
- Features: $X = [X_0, X_1, \cdots, X_n]$ (words)
- Target: $y \in \{spam, ham\}$
- Assumptions
  - $X_i \not\perp\!\!\!\perp y \; \forall X_i \in X$
  - $X_i \perp\!\!\!\perp X_j \mid y \; \forall X_i, X_j \in X$ (“naïve”)
    - all features are **independent** of each other (which isn't always true in real life, but works surprisingly well in practice)

Structure and Semantics

- Simplified BN
  - Class y is the parent of all feature nodes
  - Find the most likely y from X
  - $\hat{y} = argmax_{y \in Y} P(Y = y | X_0 = x_0, X_1 = x_1, \dots)$



Output
(predicted y)

Learning Objective
(target maximization quantity)

Structure and Semantics

- How to compute the learning objective in terms of BN's CPTs?

$$P(Y|X_0, X_1, \dots) = \frac{P(Y, X_0, X_1, \dots)}{P(X_0, X_1, \dots)}$$

Conditioning Rule

$$= \frac{P(X_0, X_1, \dots|Y) * P(Y)}{P(X_0, X_1, \dots)}$$

Conditioning Rule

$$= \frac{P(X_0|Y)P(X_1|Y) \dots P(Y)}{P(X_0, X_1, \dots)}$$

Independence

Which part matters?

$$P(X_0|ham)P(X_1|ham) \dots P(ham) \; ? \; P(X_0|spam)P(X_1|spam) \dots P(spam)$$

Decision Rule

- $\hat{y} = argmax_{y \in Y}\, P(Y = y | X_0 = x_0, X_1 = x_1, \dots)$

  $\propto argmax_{y \in Y}\, P(x_0|y)P(x_1|y) \dots P(y)$

  $= argmax_{y \in Y} P(y) \prod_{x_i} P(x_i|y)$

Practical Computing Problem: numerical underflow -- # of features is large

- $\log(A \cdot B) = \log(A) + \log(B)$
- $argmax_{y \in Y} P(y) \prod_{x_i} P(x_i|y) = argmax_{y \in Y} \log\big(P(y)\big) + \sum_{x_i} P(x_i|y)$

Example

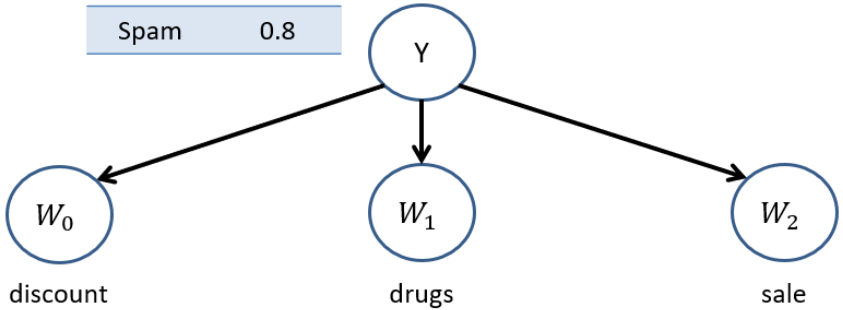
$$argmax_{y \in Y} P(y) \prod_{x_i} P(x_i|y)$$

- Classifying the message "discount drugs sale"

$$P(spam) * P(drugs|spam) * P(sales|spam) * P(discount|spam) = 0.0192$$

$$P(ham) * P(drugs|ham) * P(sales|ham) * P(discount|ham) = 0.0024$$

Y	P(Y)
Spam	0.8



W_i	P(W_i | Y=Spam)
"drugs"	0.3
"sale"	0.4
"discount"	0.2
…	…

W_i	P(W_i | Y=Ham)
"drugs"	0.2
"sale"	0.3
"discount"	0.2
…	…